Chapter C2: Chemical patterns

Overview

This chapter features a central theme of modern chemistry: it traces the development of ideas about the structure of the atom and the arrangement of elements in the modern Periodic Table. Both stories show how scientific theories develop as new evidence is made available that either supports or contradicts current ideas.

Atomic structure is used to help explain the behaviour of the elements. Trends and patterns shown by the physical and chemical properties in groups and in the transition metals are studied.

The first two topics of the chapter give opportunities for learners to develop understanding of ideas about science; how scientific knowledge develops, the relationship between evidence and explanations, and how the scientific community responds to new ideas. The later topics present some of the most important models which underpin an understanding of atoms, chemical behaviour and patterns and how reactions are represented in chemical equations.

Topic C2.1 looks at the development of ideas about the atom and introduces the modern model for atomic structure, including electron arrangements. Topic C2.2 considers the development of the modern Periodic Table and the patterns that exist within it, focusing on Groups 1 and 7, with some reference to Group 0. Topic C2.3 focuses on extending an understanding of atomic structure to explain the ionic bonding between ions in ionic compound. This leads on to Topic C2.4 which studies using equations and symbols to summarise reactions**.** Finally, in Topic C2.5, separate science only content addresses the unique nature of the transition elements.

Learning about Chemical Patterns before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- know the properties of the different states of matter (solid, liquid and gas) in terms of the particle model, including gas pressure
- know changes of state in terms of the particle model
- be aware of a simple (Dalton) atomic model
- know differences between atoms, elements and compounds
- know chemical symbols and formulae for elements and compounds
- know conservation of mass in changes of state and chemical reactions
- understand chemical reactions as the rearrangement of atoms
- be able to represent chemical reactions using formulae and using equations
- know some displacement reactions
- know what catalysts do
- be aware of the principles underpinning the Mendeleev Periodic Table
- know some ideas about the Periodic Table: periods and groups; metals and non-metals
- know how some patterns in reactions can be predicted with reference to the Periodic Table
- know some properties of metals and non-metals.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.